Navigation from the C-arm alone: naming the vertebral level on any fluoroscope with a physics-parameterised synthetic corpus

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Conflicts of interest: the author has a commercial interest in a level-confirmation product under development.

Data and code: every dataset is public and listed in Table 1 with its licence; the data manifest (file-list hashes), the claims ledger that ties every number here to its result file, and the reproduction chain are released with the code.

Abstract

Purpose: To identify the vertebral level from operating-room C-arm exposures alone, without a preoperative 3-D scan or a tracked instrument, on C-arm models the model never saw.

Materials and Methods: Retrospective study of public data (Table 1). A level model developed on plain radiographs counted from the sacrum across overlapping exposures. For each C-arm with paired real frames and CT, nine physical terms of image formation were fitted leave-one-group-out and exposure was modelled as a sampled distribution; whole-spine CTs were rendered under each C-arm's real poses and styled through the fitted terms. Detectors were trained with one C-arm held out and scored on its real frames, with case-level bootstrap intervals.

Results: On 148 intraoperative lumbar cases with cone-beam CT truth, whole lateral frames named 98.2 % of 617 vertebrae exactly [95.7, 100] and 92.9 % with three in the field; without that device's frames the model named 96.3 % of lateral but 37.9 % of AP vertebrae. A detector trained on two other C-arms found 90.6 % of the held-out one's vertebrae under a matched box convention. Nine fitted terms explained held-out real frames at R^2 0.905 (Siemens Cios Fusion) and 0.952 (Philips BV Pulsera); real C-arms occupied separable regions of image-statistics space. [[C7/C8] held-out device results of the styled-corpus detector and the scaling with the number of training devices.]

Conclusion: [Written when C7 is read: whether one detector trained on physics-styled renders of $k-1$ C-arms reaches counting grade on the kth.]

Summary statement: [one sentence, from C7.]

Key points

- Level naming on real intraoperative lateral frames reached 98.2 % exact with CBCT truth and no preoperative scan; a plain-film model transferred to laterals (96.3 %) but not to AP frames (37.9 %).
- Each C-arm is a separable image domain; nine fitted physical terms reproduce its frames (R^2 0.905–0.952 held out) and a sampled exposure distribution covers 80.2 % of unseen frames on the intensifier.
- [[C7] the pooled-device result.]

Introduction

A vertebral level is not marked; it is counted. With navigation, the level is read off a registration to an intraoperative cone-beam CT or 3-D fluoroscopic scan [1]. Without it, the surgeon counts on fluoroscopy from a known landmark, up from the sacrum or down from C1 or C2, across successive exposures [1, 2], and wrong-level surgery persists despite the count [3]. A lumbar needle localisation costs about three exposures, 4 s of fluoroscopy and 2 minutes [4], and the level first exposed is wrong in 4.3 % (13 of 301) to 15 % (15 of 100) of lumbar operations before the check film corrects it [5, 6]. Half of responding spine surgeons report at least one wrong-level operation [7, 8]; a registry of 34,544 degenerative operations recorded 0.12 %, and 0.42 % in the thoracic spine [9].

Avoiding the count is paid for. One centre, Mayo Clinic in Rochester, placed 179 preoperative gold fiducials over 38 months, 176 thoracic, at 51.9 minutes and 183 mGy each, to let the surgeon find the level [10]—about $1,750–1,950 of staffed procedural time apiece at $34–37 per operating-room minute [11]; implanting one was costed at $1,237.55 (fluoroscopic) to $1,267.03 (CT-guided) under 2013 procedural coding [12]. Navigation removes the count, but not for everyone: cone-beam CT navigation becomes cost-saving only above 254 instrumented procedures a year [13]; at an annualised $414,634, 61–86 % of United States inpatient facilities cannot justify it [14]. Departmental use at least monthly is 69.5 % in high-income countries against 24.1 % in low- and middle-income countries [15], and among African spine surgeons 96.5 % have fluoroscopy but 8.8 % have navigation with intraoperative CT [16]. LevelCheck, the clinically translated decision-support tool for fluoroscopic level checking, registers the radiograph to a preoperative CT or MR [2, 17, 18]. Every method that reports naming accuracy on clinical intraoperative images takes the level from another image of the same patient: by 3-D–2-D registration to a preoperative CT or MR, or by matching vertebrae to a preoperative radiograph, which named 73.3 % of vertebrae on 31 C-arm images [19]. Published per-vertebra detection recalls on C-arm images (0.96–0.98) name nothing [20], and registration-free naming has been reported only on cadaver long-length tomosynthesis, at 88.3 % [21]. Every hospital has the C-arm.

We ask whether the navigation can be built from the C-arm shots alone. Two obstacles are measured here rather than assumed: a model that names vertebrae on plain radiographs does not carry to fluoroscopy, and every C-arm is its own image domain, so a model that works on one fluoroscope fails on the next. We parameterise each C-arm's image formation with physical terms fitted on paired real frames, sample each device's exposure distribution, and train one detector on renders styled as every device but the one under test,

scored on that device's real frames.

Materials and Methods

Data. Table 1 lists every source: five C-arms with paired real frames, CT and pose (a mobile flat-panel C-arm, a mobile image intensifier, two linac-mounted kV imagers, a fixed flat-panel turntable), intraoperative lumbar frames with cone-beam CT truth, six further C-arms imaging one sawbone phantom, literature fluoroscopy, and five plain-film corpora. Splits are by case; no test case of any device is in any training list (audited by script).

Level naming. A ResNet-18 encoder over fixed-scale vertebra crops with a recurrent head over 2–9 vertebrae, trained on plain films and renders [22, 23]; on fluoroscopy, overlapping exposures are linked into a chain and counted from the sacrum (Figure 1a–c). Truth on the intraoperative cases is the cone-beam CT registered to each frame and labelled by two independent segmentation tools; only levels they agree on are scored.

Confirmation. One exposure holding the instrument is matched onto the panorama by normalised cross-correlation over a small scale range, and the body under the tip takes its node's ordinal; a match is kept only when its peak stands above the best peak more than half a body away, the alias a periodic column invites. This was run on eighteen CTSpinoPelvic1K spines carrying a sixth lumbar vertebra, three seeds each, with the tip placed by walking in from posterior to the first bone it meets.

Image formation per C-arm. For each device with paired data, the projected water and bone paths of the registered CT are pushed through a polyenergetic transmission, a scatter pedestal, a collimator penumbra, a quantum draw, receptor blur, edge enhancement, vignetting and heel terms, and a per-frame linear display match; the nine terms are fitted leave-one-group-out (specimen, case or view) and each has a physics default beside it. Exposure-dependent terms are then estimated per frame with only the quantum count free, and stored as a Gaussian copula conditioned on water-equivalent thickness, so that a whole frame can be drawn. Coverage is the share of held-out frames whose terms are typical under the sampler.

The synthetic corpus and the held-out device test. Whole-spine CTs are placed under each C-arm's real poses, ray-cast once at the device's geometry, and styled by draws from a device's card. A detector (YOLO11m from COCO) is trained on styled renders of every device but one, mixed with the real frames of the training devices, and scored on the held-out device's real test frames with the box convention matched to that device's truth. [[C8] repeated with one and two training devices; [C9] device-specific fine-tuning.]

Statistics. Case-level bootstrap 95 % intervals; paired bootstrap for differences; pass criteria stated before each experiment was read; every quoted number is regenerated from its result file by a ledger check that fails on drift.

Results

Plain film to fluoroscopy. Renders alone do not make a detector. Trained on renders with no real image, a detector found 71.7 % of vertebral bodies on real radiographs (25 spines, 71 frames, 233 bodies), against 99.1 % for the same architecture trained on film; on renders, its own domain, it found 93.5 %, so it was weaker before any shift and lost more across it. Film-trained naming transferred to intraoperative laterals but not to AP frames: without C-ArmLSpine frames in training, the level model named 96.3 % of lateral vertebrae and 37.9 % of AP vertebrae exactly. On held-out real radiographs, split by patient, the level model named 99.9 % of 3,720 cervical vertebrae (CSXA, levels from surgeons' keypoints [24]) and 98.0 % of 2,698 lumbar vertebrae (BUU-LSPINE, levels assigned by position) exactly inside a random window of two to nine, and 94.9 % with three lumbar vertebrae visible. NHANES II is not reported: it entered the training bank twice, as two sources split independently, so its validation images also appear in training.

Level naming on real intraoperative frames. On 148 cases, whole lateral frames named 98.2 % of 617 scorable vertebrae in 141 cases exactly [95.7, 100], 97.0 % with five vertebrae in the field, 92.9 % with three and 72.8 % with one; AP frames 96.7 %. On whole frames no error exceeded one level in either view (100 % and 100 % within one); with a single vertebra in the field 90.2 % were within one.

Each C-arm is its own domain. A detector trained on the real frames of two device families found 90.6 % of vertebrae and 83.9 % of sacra on the third, C-ArmLSpine, whole frames, under the outline convention its training truth used (5.2 % under the body-box convention of the held-out device's own truth — a scoring

artefact, shown by the same-device control falling from 90.6 to 40.8 % under the swapped convention). Arms whose training and test truths use different box conventions (a detector trained only on C-ArmLSpine body boxes, scored on the outline boxes of COBRA and DeepFluoro) are not interpretable until rescored and are not reported. In image-statistics space (Figure 1d; 13,875 frames from 37 corpora, 7,311 of them real) each C-arm occupies its own region: the Bhattacharyya overlap between two C-arms runs from 0.005 (the Siemens Cios frames of DeepFluoro against the Philips intensifier at Geneva) to 0.675 (the Elekta imager against that same intensifier). The domains themselves do not separate that way. A classifier trained to call a frame film or fluoroscopy, with one corpus held out entirely, reached 71.1 % against a majority share of 71.8 %: it labels an unseen C-arm corpus as film. What these statistics separate is the corpus, not the modality, which is why a detector must be scored on a device held out whole rather than on a held-out domain.

Confirmation at a transitional vertebra. On eighteen CTSpinoPelvic1K spines carrying a sixth lumbar vertebra, three seeds each (54 runs on clean renders), the chain had exactly one node per body in 45 and the instrument's own exposure, matched onto that chain, named L6 in 36; under a noise model fitted to real radiographs the same runs gave 45 and 35. The misses were not the count: two spines placed the tip one body caudal on every seed and one never matched (Figure 1f). Styling these lateral sweeps through a device card is not yet possible. Each card's exposure model is fitted over 9–20 cm of water-equivalent thickness; a lateral abdomen is 28–38 cm, where the fitted photon count leaves about a dozen quanta per pixel and the bone's 19 % primary modulation sits under the quantum noise. Scatter is not the term that hides it: removing it changes the grey-level spread over the vertebral body by a few levels, while the noiseless render of the same frame shows the bodies and the ribs.

Nine physical terms reproduce each C-arm. Held-out R^2 rose from 0.899 to 0.905 on the Cios Fusion and from 0.917 to 0.952 on the BV Pulsera (Elekta XVI, 0.964 to 0.974). Physics defaults sat within the fold spread for beam hardening, scatter and vignetting on the intensifier. With only the quantum count free per frame, the Cios card gives a median of 5,943 photons per 0.78-mm pixel; the sampler covered 80.2 % of held-out intensifier frames and 84.8 % of held-out kV-CBCT frames, but 10.1 % of held-out cadavers on the Cios, where each specimen was its own exposure regime.

One detector for every C-arm. [[C7] designs A and B, three held-out devices, seeds; [C8] scaling curve; the pre-stated pass criterion.]

Device-specific fine-tuning; classes; benchmark. [[C9]–[C11].]

Discussion

A level can be named on operating-room frames without a preoperative scan or a tracker: on real intraoperative laterals the naming was exact in 98.2 % of vertebrae, and the instrument's own exposure carried that name back to the bone it touched. [[C7] one sentence: whether a detector trained on physics-styled renders of $k-1$ C-arms reaches counting grade on the kth, and what that says about buying a new device with a parameter fit instead of a labelled dataset.]

What is not shown matters as much. The physics-styled frames are not yet statistically inside the real domain on pooled image statistics, so the detector's score on a held-out device, not that comparison, is the test that counts. One device has intraoperative truth; the others are cadaver, phantom or non-spine, and no C2 anchor has been demonstrated on a real intraoperative frame. Every sweep number here uses the annotation's boxes, which measures the linker and not a deployable system, and the executed move is the commanded move by construction, so the exactness figures are an upper bound on what the search window achieves when a radiographer's move is not the commanded one; no axis-resolved figure for a pure cranio-caudal pan has been published to bound it. In the transitional-vertebra runs the count was right more often than the instrument's assignment, which failed on two spines on every seed. The device cards cannot yet render a 28–38 cm lateral abdomen, so the styled version of that experiment is not reported. Comparison with prior art is indirect, because no published system names vertebrae from the content of real intraoperative fluoroscopy above 73.3 % [19], and that figure imports identity from another image of the same patient.

The next tests are a phantom on a real C-arm, a cadaver, and a human series, in that order, each with the count scored against cone-beam CT truth rather than against a render.

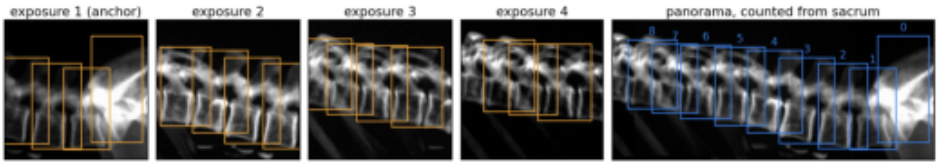
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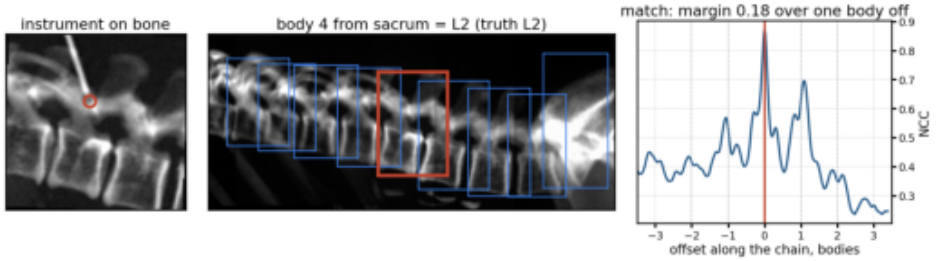
(a) Processing chain



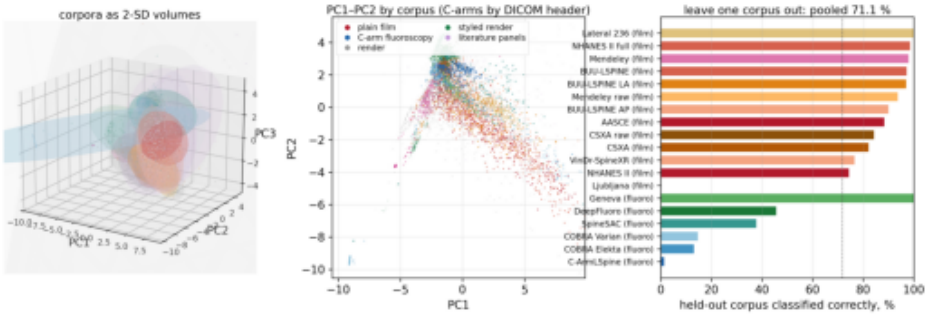
(b) Sweep on a prone patient (head left, posterior up), counted from the sacrum



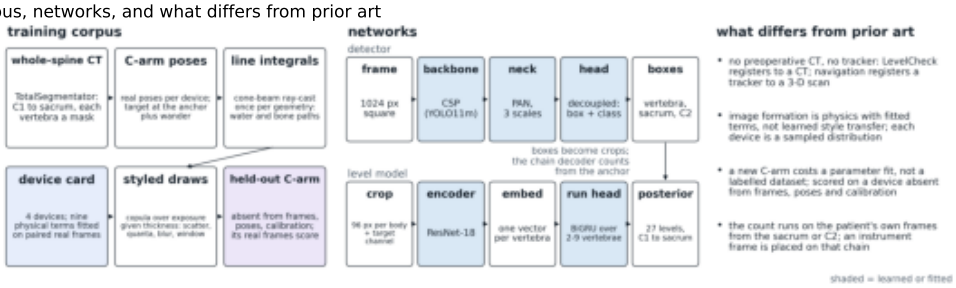
(c) Confirmation: instrument on bone, matched onto the chain



(d) Every corpus in one image-statistics space



(e) Training corpus, networks, and what differs from prior art



(f) Confirmation at a sixth lumbar vertebra, counted from the sacrum (synthetic frames, film noise model)

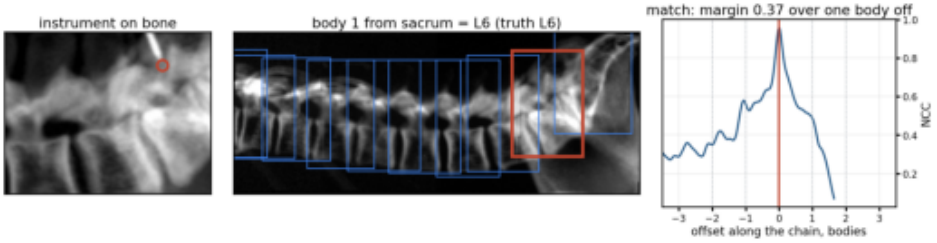


Figure 1: (a) The chain. (b) A sweep on a prone patient, head left, posterior up, counted from the sacrum. (c) Confirmation: an instrument stops on bone, its exposure is matched onto the chain, and the match is drawn against offset in body lengths. (d) Every corpus in one image-statistics space. (e) The corpus and the networks. (f) The same at a sixth lumbar vertebra.

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Table 1: X-ray datasets. Generated from the data manifest, the device cards and the header census (tools/rsnai_table1.py).

Dataset	C-arm / detector	Pitch (mm)	Stored	kVp	Frames / subjects	Paired with	Licence
DeepFluoro	Siemens Cios Fusion (csi fpd)	0.194	log display	—	366 frames, 6 cadaveric pelves	CT + C-arm pose per frame	CC BY-NC-SA 4.0
C-ArmLSpine	surgio h51	0.15 (307.2 mm / 2048; header PixelSpacing is a 1.0 placeholder)		73–111	296 AP + lateral frames, 148 cases (+ 2 raw sweeps)	CBCT per case; pose by registration	CC BY 4.0
COBRA2026, Elekta	Elekta XVI kV imager (four European centres A-D) (unknown fpd)	0.8	linear	120–120	projection sweeps, 152 cases downloaded of 692 public	planning CT (deformably registered) + RTK geometry	CC BY-NC 4.0
COBRA2026, Varian	—	—	—	—	24 cases downloaded (centres F, G)	planning CT + RTK geometry	CC BY-NC 4.0
Geneva hip	Philips BV Pulsera (ii)	0.2904	linear	65–90	116 frames, 19 poses, resin phantom	CT + gold-standard pose	CC BY-SA 4.0 (repository)
Ljubljana 2D-3D-GS	Trixell (Ljubljana 2D-3D-GS) PIXIUM 4600 flat panel, fixed radiograph on a turntable (not a C-arm) (csi fpd)	0.286	linear	—	18 frames, cadaver L1–L5	CT + gold-standard pose	CC BY-NC-ND
SpineSAC	—	—	—	—	raw frames of a sawbone phantom on six C-arms	phantom CT + shipped poses; no bone truth yet	—
Plain-film corpora	—	—	—	—	BUU, CSXA, NHANES, Mendeley, AASCE	keypoint / level labels	per source
FluoroCap panels	—	—	—	—	29,484 literature fluoroscopy panels (unpaired)	none	figure reuse only